

Feedback-Dressed Dynamic Topology of Inverter-Dominated Power Systems: Hidden Cycles, Reinforcement Reversals, and Controller-Aware Screening

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Abstract—Conventional grid-strength screens describe the physical electrical network, whereas inverter-rich stability is determined by a closed-loop network that includes controller dynamics. This paper develops a controller-aware interpretation of that discrepancy. A controller self-energy $\Sigma(s)$ is composed with the frequency-dependent network resolvent $G_r(s)$ to form a *feedback-dressed network operator* $K_H(s) = G_r(s)\Sigma(s)$. The corresponding controller-side return ratio $R_c(s)$ has the same nonzero loop spectrum, and the full characteristic determinant factors exactly into local network dynamics, local controller dynamics, and an interaction determinant. Schur condensation then exposes frequency-dependent effective edges and cycles that need not exist in the retained physical graph. The framework is evaluated on a transparent IEEE 39-bus proof-of-concept: a MATPOWER AC operating point is coupled to a stylized grid-following converter at bus 34. Reinforcing branch 19–20 by $1.8\times$ improves algebraic connectivity by 0.694%, reduces a bus-34 network-impedance proxy by 4.60%, and reduces AC losses by 0.227%, yet collapses the network–controller feedback margin by 91.3% and reduces the controller-family damping ratio from 2.89% to 0.186%. A passive reduced cycle gain of 0.00596 becomes 1.013 after controller dressing and falls to 0.0787 after controller retuning while the physical reinforcement is retained. An exhaustive 46-branch action screen, frozen before nonlinear simulation, ranks three deceptive reinforcements whose nonlinear controller-stress amplification is ordered exactly as predicted. Fiedler support and the hidden dynamic red zone show low spatial overlap. The results suggest that controller-aware planning should screen not only physical strength but also the dynamic feedback topology created by network propagation and controller self-energy.

Index Terms—inverter-based resources, grid-following inverter, small-signal stability, graph signal processing, Fiedler vector, Schur complement, self-energy, hidden dynamic graph, feedback cycle, transmission planning.

I. INTRODUCTION

LOW-INERTIA and inverter-dominated grids are often screened with quantities derived from the physical network: short-circuit strength, impedance, loading, algebraic connectivity, Laplacian eigenvectors, or reduced L – M – D models. These quantities are useful because they summarize the network seen by power-flow or electromechanical coordinates. They are not, however, a complete representation of the

feedback paths created by phase-locked loops (PLLs), current controllers, outer loops, exciters, governors, stabilizers, and other dynamic devices.

Impedance-based stability criteria already establish that grid and converter dynamics must be analyzed jointly [2], and interaction-admittance models extend that view to multi-inverter networks [3]. PLL–grid coupling is also known to change small-signal stability and can require controller-side remedies [4]. Separately, Kron reduction formalizes how eliminating electrical nodes creates effective edges in reduced networks [5], and graph signal processing (GSP) provides a spectral language for signals and operators on graph modes [7]. The question pursued here is narrower: *how should a physical topology action be mapped to the closed-loop dynamic feedback that actually determines an inverter-rich stability margin?*

The key observation is that controller dynamics are not stability-relevant merely because they exist. Their effect must be propagated through the dynamic susceptibility of the network and returned through the feedback loop. We therefore distinguish four objects:

$$L_P \longrightarrow \Sigma(s) \longrightarrow K_H(s) = G_r(s)\Sigma(s) \longrightarrow R_c(s), \quad (1)$$

where L_P describes the physical graph, $\Sigma(s)$ is the controller self-energy on retained network states, $G_r(s)$ is the network resolvent, $K_H(s)$ is the feedback-dressed network operator, and $R_c(s)$ is the controller-side return ratio. The proposed *hidden margin* is based on the distance of $I - R_c(j\omega)$ from singularity.

The algebra used below—Schur complements, determinant factorizations, Woodbury updates, and spectral graph projections—is standard. The proposed contribution is their specialization into a single power-system workflow from *physical action* to *hidden feedback topology* to *dynamic consequence*. The main contributions are:

- an exact determinant factorization that separates local network and controller dynamics from interaction-induced modes;
- a feedback-dressed effective graph whose hidden edges and cycles are generated by Schur condensation and controller/network propagation;

- a low-rank topology-to-feedback map for transmission actions, together with a controller-side hidden feedback margin;
- a GSP interpretation showing why a Fiedler-only or raw- Σ view is insufficient, while feedback-dressed cross-mode coupling is more informative;
- an exhaustive IEEE 39-bus proof-of-concept in which electrically beneficial reinforcements are ranked before nonlinear simulation and the frozen ranking predicts nonlinear controller stress.

The paper intentionally does not claim independent DAE/EMT validation. MATPOWER [8] is used for the AC operating point and physical-network metrics; ANDES [9] is identified as the next independent validation environment.

II. PHYSICAL AND DYNAMIC NETWORK MODELS

A. Physical AC graph and Fiedler coordinates

Let a solved AC operating point be denoted by $x^* = (\theta^*, V^*, \dots)$. A local physical coupling matrix can be formed from the angle Jacobian or, in the lossless approximation,

$$L_P(x^*) = B W(x^*) B^\top, \quad W_{ee} = b_e V_i^* V_j^* \cos(\theta_i^* - \theta_j^*), \quad (2)$$

where B is the branch incidence matrix. Its ordered eigenpairs satisfy

$$L_P U = U \Lambda, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots, \quad (3)$$

with λ_2 the algebraic connectivity and u_2 the Fiedler vector [6]. The graph Fourier transform of a bus signal y is $\hat{y} = U^H y$ [7]. These coordinates provide a natural baseline for asking whether controller dynamics preserve or mix the physical graph modes.

B. Linearized DAE and controller self-energy

Consider a linearized dynamic model after algebraic elimination,

$$\dot{x} = A x, \quad A = \begin{bmatrix} A_{rr} & A_{rc} \\ A_{cr} & A_{cc} \end{bmatrix}, \quad (4)$$

where r denotes retained network/interface states and c controller/device states. Define

$$T_r(s) = sI - A_{rr}, \quad T_c(s) = sI - A_{cc}, \quad (5)$$

and the network-side controller self-energy

$$\Sigma_r(s) = A_{rc} T_c(s)^{-1} A_{cr}. \quad (6)$$

The exact retained operator is

$$T_{\text{eff},r}(s) = T_r(s) - \Sigma_r(s). \quad (7)$$

For electromechanical retained coordinates, (7) contains the familiar second-order form $s^2 M + s \Sigma_D(s) + L$ as a special representation, but the state-space form is used here because it does not require controller dynamics to be collapsed into a scalar damping coefficient.

C. Feedback-dressed network operator and hidden margin

Let

$$G_r(s) = T_r(s)^{-1}, \quad G_c(s) = T_c(s)^{-1}. \quad (8)$$

We define the feedback-dressed network operator

$$K_H(s) = G_r(s) \Sigma_r(s) = G_r A_{rc} G_c A_{cr} \quad (9)$$

and the controller-side return ratio

$$R_c(s) = G_c(s) A_{cr} G_r(s) A_{rc}. \quad (10)$$

The proposed hidden feedback margin is

$$d_H = \min_{\omega \in \Omega} \sigma_{\min}[I - R_c(j\omega)]. \quad (11)$$

Unlike a physical strength index, d_H is a closed-loop quantity: it becomes small when a network-mediated controller return approaches singular feedback closure.

III. ALGEBRAIC STRUCTURE OF HIDDEN DYNAMIC TOPOLOGY

A. General interaction determinant

Proposition 1 (Interaction determinant): For frequencies where $T_r(s)$ and $T_c(s)$ are nonsingular,

$$\det(sI - A) = \det T_r(s) \det T_c(s) \det[I - R_c(s)]. \quad (12)$$

Equivalently,

$$\det(sI - A) = \det T_c(s) \det T_r(s) \det[I - K_H(s)]. \quad (13)$$

Proof: Apply the Schur determinant identity to $sI - A$ using either diagonal block. The second equality follows from Sylvester's determinant theorem $\det(I - XY) = \det(I - YX)$. ■

Equation (12) separates local poles of the network and controller blocks from interaction-induced zeros of $\det(I - R_c)$. This is the precise sense in which the proposed hidden topology describes dynamics that are not visible from the physical Laplacian alone.

Proposition 2 (Network/controller spectral equivalence): $K_H(s)$ and $R_c(s)$ have identical nonzero eigenvalues, including algebraic multiplicities.

Proof: Write $X = G_r A_{rc}$ and $Y = G_c A_{cr}$. Then $K_H = XY$ and $R_c = YX$. The nonzero spectra of XY and YX coincide. ■

This result justifies analyzing the interaction in either network coordinates (useful for GSP and localization) or controller coordinates (useful for low-dimensional loop screening).

B. Hidden effective edges and cycles

Partition a dynamic operator into retained ports b and hidden states h :

$$T(s) = \begin{bmatrix} T_{bb} & T_{bh} \\ T_{hb} & T_{hh} \end{bmatrix}. \quad (14)$$

Eliminating h gives

$$T_{\text{eff}}(s) = T_{bb} - T_{bh} T_{hh}^{-1} T_{hb}. \quad (15)$$

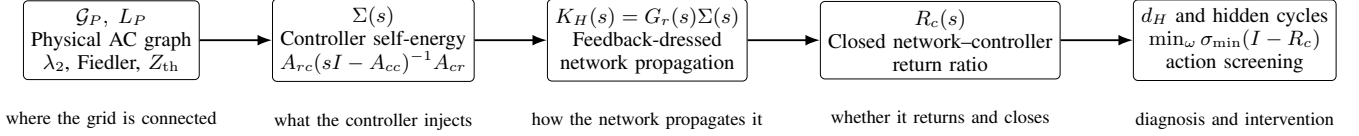


Fig. 1. Hierarchy used in this paper. The physical Laplacian is a structural description; controller self-energy becomes stability-relevant after propagation through the network resolvent and closure of the network–controller loop.

An off-diagonal element of T_{eff} can be nonzero even when the corresponding retained ports have no direct physical edge. Such an element is called an *effective edge*; if it changes only after controller dressing it is controller-mediated rather than a purely passive Kron edge.

For a three-port normalized operator $T_{\text{eff}} = D(I - W)$ with $\text{diag}(W) = 0$, direct determinant expansion gives

$$\det(I - W) = 1 - W_{12}W_{21} - W_{13}W_{31} - W_{23}W_{32} - W_{12}W_{23}W_{31} - W_{13}W_{32}W_{21}. \quad (16)$$

Thus the interaction determinant is exactly decomposed into three directed 2-cycles and two oriented 3-cycles. For larger scalar-port graphs, the Leibniz expansion yields signed collections of vertex-disjoint cycles, while the log-determinant trace expansion organizes closed walks.

C. Rank-one cycle closure and topology updates

When $A_{rc} = bq^T$ and $A_{cr} = uc^T$, the return ratio has rank at most one. Its only possible nonzero eigenvalue is

$$h(s) = \underbrace{q^T G_c(s) u}_{G_c^{\text{loop}}(s)} \underbrace{c^T G_r(s) b}_{G_n^{\text{loop}}(s)}, \quad (17)$$

and

$$\det(sI - A) = \det T_r \det T_c [1 - h(s)]. \quad (18)$$

Hence an interaction-induced mode satisfies $h(s) = 1$ when s is not already a local pole.

At a frozen operating point, let a physical action update the network operator by rank one,

$$T_r(p) = T_r + p u_e v_e^T. \quad (19)$$

Sherman–Morrison gives

$$G_r(p) = G_r - \frac{p G_r u_e v_e^T G_r}{1 + p v_e^T G_r u_e}. \quad (20)$$

Substitution into (17) yields the exact finite-action map

$$h(p, s) = h(0, s) - \frac{p G_c^{\text{loop}}(s) [c^T G_r u_e] [v_e^T G_r b]}{1 + p v_e^T G_r u_e}. \quad (21)$$

For a full AC action, (21) is the direct frozen-point contribution; the total derivative also contains the operating-point shift from the implicit-function theorem.

D. GSP view of controller-mediated mode mixing

Let U be the physical Laplacian basis. A raw controller-dressing map is

$$\hat{\Sigma}(j\omega) = U^H \Sigma_{\text{bus}}(j\omega) U. \quad (22)$$

Its off-diagonal terms quantify controller-induced mixing between physical graph modes. The experiments below show that raw $\hat{\Sigma}$ is not sufficient. The more informative object is the feedback-dressed projection

$$\hat{K}_H(j\omega) = U^H K_{H,\text{bus}}(j\omega) U, \quad (23)$$

which includes both controller dressing and frequency-dependent network propagation. We use

$$g_{\max} = \max_{k \neq \ell} |[\hat{K}_H]_{k\ell}| \quad (24)$$

as an interpretable cross-mode diagnostic rather than as a replacement for the exact hidden margin (11).

IV. IEEE 39-BUS NUMERICAL CASE AND FALSIFICATION PROTOCOL

A. Exact numerical model and reproducibility

The physical case is the standard MATPOWER IEEE 39-bus system. The AC power flow is re-solved after every topology action. A Newton–Raphson implementation from the published case data reproduces the stored MATPOWER solution with maximum voltage error 4.80×10^{-8} p.u., maximum angle error 4.95×10^{-7} degrees, and infinity-norm residual 1.43×10^{-13} p.u. This establishes that the physical-network metrics below are evaluated on the intended IEEE 39-bus operating point rather than on a separate approximate load-flow model.

The star numerical case is shown in Fig. 2. Bus 20 carries 680 MW and 103 Mvar in the MATPOWER case. Branch 19–20 has $r = 7 \times 10^{-4}$ p.u., $x = 0.0138$ p.u., and tap 1.06; bus 34 is connected to bus 20 through a branch with $r = 9 \times 10^{-4}$ p.u., $x = 0.0180$ p.u., and tap 1.009. In the dynamic proof-of-concept the synchronous unit at bus 34 is replaced by one stylized grid-following converter. The controller is intentionally transparent:

$$\dot{\phi} = \omega_{\text{pll}}, \quad (25)$$

$$\dot{\omega}_{\text{pll}} = -2\zeta_{\text{pll}}\omega_n\omega_{\text{pll}} - \omega_n^2 \sin(\phi - \theta_{34}), \quad (26)$$

$$\tau \dot{p}_c = -p_c + \text{sat}_{\bar{p}} [-K_e \sin(\phi - \theta_{34}) - K_\omega \omega_{\text{pll}} + u_d]. \quad (27)$$

The nominal parameters are $\omega_n = 17.9016$ rad/s, $\zeta_{\text{pll}} = 0.7874$, $K_e = 98.220$, $K_\omega = 1.65$, $\tau = 0.04956$ s, network damping coefficient $D_g = 2.8324$, and nonlinear limit $\bar{p} = 0.6$

p.u. The retuned case changes only K_ω from 1.65 to 2.20. These parameters are a stress-test surrogate, not a vendor-calibrated GFL model; critically, they are fixed before the 46-action atlas is generated.

B. Numerical walk-through: safe $\rightarrow$ near closure $\rightarrow$ unstable $\rightarrow$ repaired

The central case is a continuous reinforcement of branch 19–20, parameterized by the admittance multiplier α . Table I gives the full numerical ledger. The sequence is deliberately explicit. At $\alpha = 1$ the controller-family mode is stable with 2.89% damping. At $\alpha = 1.8$, all three AC/network metrics improve, but the hidden feedback margin falls from 2.87×10^{-2} to approximately 2.05×10^{-3} and the controller-family damping falls to 0.186%. Near $\alpha = 1.925$ the pole crosses the imaginary axis. At $\alpha = 2$, corresponding to an identical parallel-admittance equivalent in this surrogate, $\Re\{\lambda_c\} = 9.67 \times 10^{-3} > 0$. Finally, restoring only controller damping at the same $\alpha = 1.8$ moves the critical controller pole to $-0.502 + j9.282$ rad/s and expands the hidden margin to 5.89×10^{-2} while leaving λ_2 , $Z_{th,34}$, and AC losses unchanged from the reinforced network.

Figure 3 places all six consequences on one numerical plate. The physical graph improves monotonically; the hidden margin and rank-one loop distance collapse; the controller-family pole moves through the imaginary axis; the hidden cycle peaks around the same frequency; removing controller feedback or retuning the controller suppresses the cycle; and the independent nonlinear stress test shows the same direction of risk.

C. Exact algebraic and numerical consistency checks

The proposed interpretation is not based on a fitted surrogate index. Several identities are checked independently at machine precision. For a frozen line update the network-operator perturbation is rank one. Direct inversion and Sherman–Morrison agree to relative error 4.84×10^{-16} ; the direct and updated return ratios agree to 1.18×10^{-16} ; and the characteristic determinant factorization in (12) agrees to 2.05×10^{-16} . Together with the AC power-flow residual above, these checks isolate modeling assumptions from algebraic/numerical error.

D. Exhaustive action atlas and blind nonlinear cross-check

A single counterexample would not justify a planning method. Therefore all 46 existing IEEE 39-bus branches are independently reinforced by $1.8\times$ and the AC operating point is re-solved for each action. The dynamic screen is then computed before nonlinear simulation. The three strongest predicted margin collapses are frozen as 19–20, 19–33, and 16–19; three electrically beneficial near-neutral controls are frozen as 8–9, 29–38, and 4–5. No re-ranking is allowed after the nonlinear trajectories are generated.

The dangerous actions increase median nonlinear controller-action RMS by 52.9%, 32.5%, and 10.1%, whereas the benign controls remain at approximately zero change. The six-action rank correlation between predicted margin collapse and

nonlinear RMS amplification is $\rho = 1.00$; event probability gives $\rho = 0.941$ ($p = 0.0051$). Halving the integration step preserves the dangerous/benign ordering. Figure 4 also shows why the result is not a Fiedler re-labeling: physical λ_2 and a Fiedler-only feedback projection are nearly uncorrelated with the hidden margin.

E. What Fiedler, GSP, self-energy, and loop closure each contribute

The exhaustive experiment produces an informative negative result. Across the 46 line actions,

$$\rho(\Delta\lambda_2, d_H) = 0.066, \quad (28)$$

$$\rho(\text{Fiedler-only projection}, d_H) = 0.058, \quad (29)$$

$$\rho(\text{raw } U^H \Sigma U, d_H) = 0.275, \quad (30)$$

$$\rho(g_{\max}(U^H K_H U), d_H) = -0.415, \quad p = 0.0041, \quad (31)$$

$$\rho(\rho(R_c), d_H) = -0.865, \quad p = 9.38 \times 10^{-15}. \quad (32)$$

Thus the missing information is not another static Laplacian statistic and not controller self-energy in isolation. The controller dressing becomes predictive only after it is propagated by the network resolvent and closed through the feedback loop. This gives the hierarchy

$$L_P \rightarrow \Sigma(s) \rightarrow G_r(s)\Sigma(s) \rightarrow R_c(s) \rightarrow d_H. \quad (33)$$

Spatially, the distinction is equally sharp. For 19–20 at $1.8\times$, the cosine overlap between Fiedler support and the critical hidden-mode support is 0.280. The critical hidden mode is concentrated at bus 34 (88.7%), bus 20 (9.0%), and bus 19 (2.0%), whereas the Fiedler vector distributes support across a broad cut structure. For 16–19 the overlap falls to 0.145. Fiedler analysis is therefore retained as the physical-structure baseline, while K_H and R_c describe the dynamic red zone and the closed-loop mechanism.

IEEE 39-bus numerical mechanism: a physical line action amplifies a hidden controller-mediated cycle

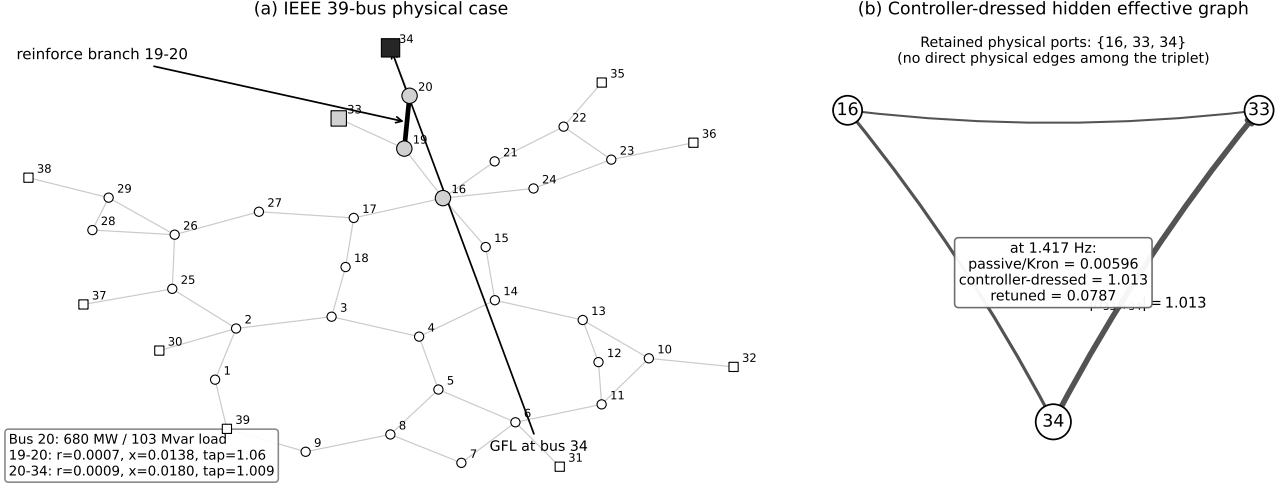


Fig. 2. IEEE 39-bus numerical mechanism. The physical action is an admittance increase on branch 19–20. Retaining only buses $\{16, 33, 34\}$ leaves no direct physical edges among the triplet, yet Schur condensation produces a controller-dressed effective graph. At 1.417 Hz the reinforced passive/Kron 33–34 cycle gain is only 0.00596, whereas controller dressing raises it to 1.013; controller retuning reduces it to 0.0787 without removing the line action.

TABLE I
NUMERICAL LEDGER FOR THE IEEE 39-BUS BRANCH 19–20 CASE. CYCLE GAIN IS EVALUATED AT THE CORRESPONDING CONTROLLER-FAMILY MODAL FREQUENCY.

Case	α	λ_2	$Z_{th,34}$	Loss (MW)	d_H	$\Re\{\lambda_c\}$	f_c (Hz)	ζ_c (%)	$ h_{33 \leftrightarrow 34} $
Base	1.000	5.3867	0.13114	43.641	2.865×10^{-2}	-0.24809	1.3665	2.888	0.134
Reinforced	1.800	5.4241	0.12511	43.542	2.047×10^{-3}	-0.01660	1.4169	0.186	1.013
Near crossing	1.925	5.4267	0.12462	43.534	6.50×10^{-4}	+0.00062	1.4225	-0.0069	1.450
Parallel-admittance equiv.	2.000	5.4281	0.12436	43.530	1.456×10^{-3}	+0.00967	1.4256	-0.108	1.747
Reinforced + retuned	1.800	5.4241	0.12511	43.542	5.886×10^{-2}	-0.50176	1.4773	5.398	0.082

TABLE II
CONSISTENCY CHECKS USED TO SEPARATE ALGEBRAIC IDENTITY FROM MODELING ASSUMPTION

Check	Value
AC PF $\ F\ _\infty$	1.43×10^{-13}
Max MATPOWER $ V $ mismatch	4.80×10^{-8} p.u.
Max MATPOWER angle mismatch	4.95×10^{-7} deg
$\text{rank}(\Delta T_r)$	1
Woodbury relative error	4.84×10^{-16}
Return-ratio update error	1.18×10^{-16}
Characteristic-factor error	2.05×10^{-16}

TABLE III
FROZEN PREDICTIONS VERSUS NONLINEAR CROSS-CHECK

Action	$\Delta\lambda_2$ (%)	Δd_H (%)	RMS amp. (%)	Event prob.
19–20	0.694	-91.3	+52.9	0.115
19–33	0.412	-62.0	+32.5	0.050
16–19	2.767	-23.9	+10.1	0.020
8–9	0.390	≈ 0	≈ 0	0.010
29–38	0.150	≈ 0	≈ 0	0.010
4–5	3.198	≈ 0	≈ 0	0.010

IEEE 39-bus numerical walk-through: stronger physical network, weaker hidden feedback, then controller-only recovery

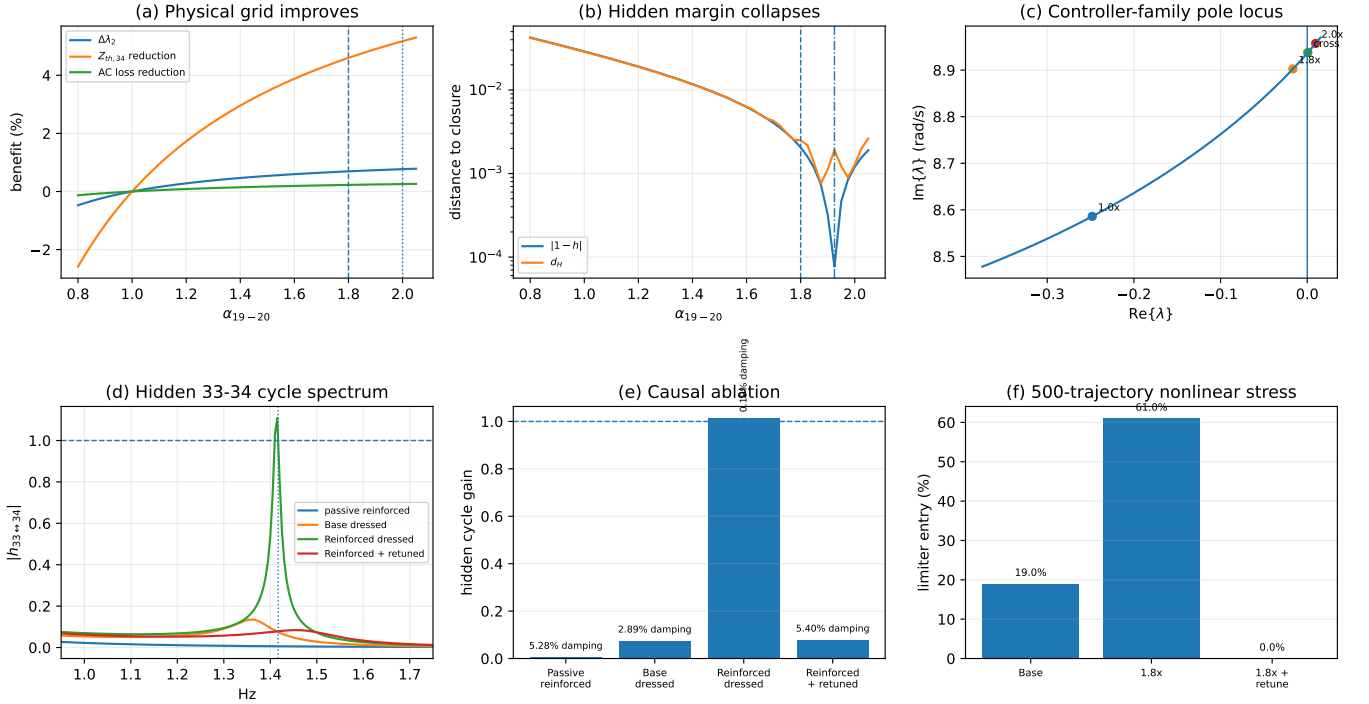


Fig. 3. Dense numerical walk-through of the branch 19–20 case. (a) Physical connectivity, Thevenin-impedance proxy, and AC losses all improve with reinforcement. (b) The feedback quantities $|1-h|$ and d_H collapse. (c) The controller-family pole crosses into the right-half plane. (d) The hidden 33–34 effective cycle is negligible in the passive reduction, becomes resonant after controller dressing, and is suppressed by retuning. (e) Causal ablation isolates the controller-mediated mechanism. (f) In a separate 500-trajectory resonance-matched nonlinear stress test, sustained current-limit entry rises from 19.0% to 61.0% after reinforcement and falls to 0/500 after retuning.

Why the result is not a single cherry-picked line: exhaustive ranking, blind nonlinear cross-check, and spatial/model hierarchy

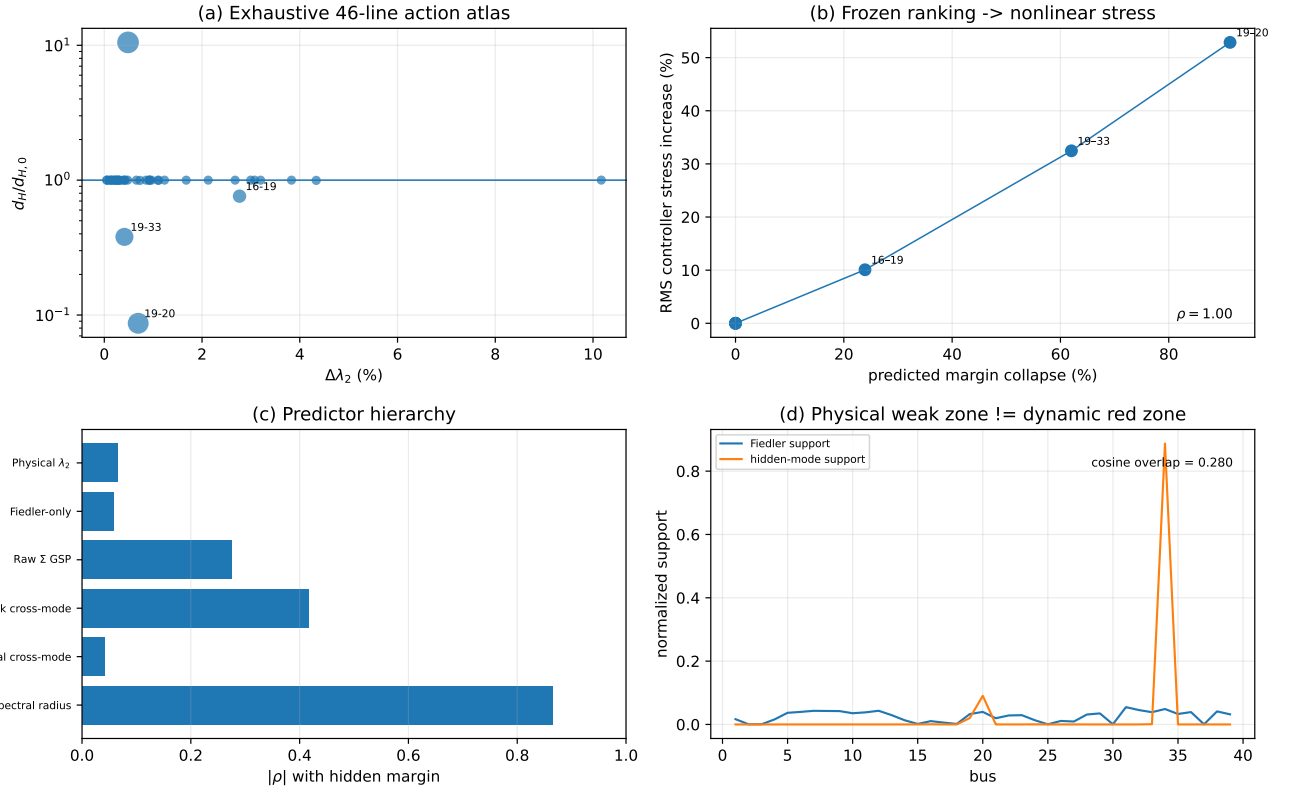


Fig. 4. Falsification-oriented screening plate. (a) Exhaustive 46-line atlas: electrically beneficial actions can have radically different feedback margins. (b) A six-action ranking frozen before nonlinear simulation predicts controller-stress amplification exactly in rank order. (c) Predictor hierarchy: physical λ_2 and Fiedler-only projections carry little information about d_H ; the full-loop spectral radius is strongest. (d) In the star case, the physical Fiedler weak-zone and critical hidden-mode support have cosine overlap only 0.280.

V. PLANNING AND INDUSTRIAL INTERPRETATION

The framework produces a decision structure that differs from a conventional weak-bus ranking. A candidate action can improve the physical network and still reduce the hidden margin. We refer to this situation as a *controller-mediated reinforcement reversal*. The operator should not necessarily reject the physical action: the causal ablation suggests a more useful workflow,

$$\text{keep reinforcement} + \text{retune limiting controller.} \quad (34)$$

This interpretation supports several industrial uses.

- 1) **Transmission screening:** use AC metrics to remove physically unattractive projects, then use d_H and hidden-cycle diagnostics to identify actions requiring detailed dynamic studies.
- 2) **Commissioning:** determine whether a topology change requires PLL/current-loop retuning before energization.
- 3) **IBR/BESS siting:** compare candidate locations by their effect on both physical strength and hidden-cycle closure rather than only SCR or Fiedler participation.
- 4) **Dynamic red zones:** project critical hidden modes and effective cycles back onto buses, lines, and devices to identify the physical support of the margin-setting feedback sector.
- 5) **Model reduction:** reduce large regions to port operators while preserving the feedback-dressed cycles that control stability rather than preserving only static terminal behavior.

In industrial facilities, the same logic can be applied to a PCC feeding active front ends, variable-frequency drives, UPS systems, filters, PV inverters, and BESS. The physical one-line diagram may be radial, yet controller/network condensation can create an effective loop among devices. The relevant engineering question becomes not only “how strong is the PCC?” but “which controller-mediated loop is closest to closure at the frequencies excited by the installed equipment?”

VI. RELATION TO PRIOR WORK

The proposed framework is complementary to impedance-based and interaction-admittance stability analysis, not a replacement. Sun’s impedance criterion [2] and subsequent multivariable approaches establish the importance of converter/grid return ratios. Lu *et al.* [3] provide a physical admittance interpretation of multi-inverter interaction, while PLL/grid coupling and controller-side remedies are demonstrated in [4]. The present work differs by making the *planning action* explicit and by tracking how that action changes a frequency-dependent effective topology, its hidden cycles, and its physical support.

Similarly, Kron reduction [5] explains why passive node elimination produces effective edges. The controller-dressed effective edges used here include that passive contribution but are separated experimentally from it. GSP [7] supplies the modal coordinates used to compare Fiedler structure with dynamic feedback mixing; the key empirical result is that raw graph modes or raw controller self-energy are not sufficient

without the network resolvent. Finally, nonlinear synchronization results in oscillator networks [10] motivate graph-based reasoning for power-system dynamics, but the present object is a frequency-dependent closed-loop operator rather than a fixed physical coupling graph.

VII. LIMITATIONS AND FALSIFIABLE NEXT TESTS

The results are intentionally framed as a proof-of-concept. The dynamic model is not a validated production GFL model; the reported stochastic event probabilities are not outage probabilities; and the controller parameters were chosen to expose the mechanism transparently. The decisive next test is independent validation with detailed DAE models.

The ranking has therefore been frozen before such validation: 19–20, 19–33, and 16–19 are the predicted dangerous actions. In ANDES, the same ranking should be tested without re-selection using REGCP1/PLL dynamics, full power-flow/DAE relinearization, and time-domain disturbances. The falsifiable outcomes are: (i) whether the frozen ranking predicts damping degradation, (ii) whether d_H predicts critical-clearing-time or current/PLL excursion, and (iii) whether controller retuning restores transient margin while the physical reinforcement remains. EMT validation should follow only for the strongest cases.

A second extension is to evaluate $K_H(j\omega, x(t))$ along nonlinear trajectories. If hidden-cycle closure consistently precedes current limiting or loss of synchronization, the framework could become a transient early-warning variable. That extension is not claimed in the present paper.

VIII. CONCLUSION

This paper argues that inverter-rich grids should be viewed through two complementary topologies: the physical topology that carries power and a frequency-dependent feedback-dressed topology that carries dynamic return. The controller self-energy $\Sigma(s)$ is necessary but not sufficient; the stability-relevant object is obtained after propagation through the network resolvent and closure of the network–controller loop. Exact determinant identities connect that object to interaction-induced modes, while Schur condensation exposes hidden effective edges and cycles.

In the IEEE 39-bus proof-of-concept, an electrically beneficial transmission reinforcement improves algebraic connectivity, network impedance, and AC losses while collapsing a controller-feedback margin and nearly eliminating damping. The critical effective cycle is negligible in the passive reduction, becomes near-unity after controller dressing, and is suppressed by controller retuning without removing the physical reinforcement. An exhaustive action screen frozen before nonlinear simulation predicts controller-stress ordering, whereas physical λ_2 , Fiedler projection, and raw self-energy mixing are substantially less informative. These results motivate a controller-aware planning workflow in which physical strength is screened first and hidden dynamic topology is then used to identify deceptive reinforcements, critical devices, and targeted control remedies.

APPENDIX A COORDINATE INVARIANCE OF HIDDEN-STATE ELIMINATION

Let the hidden coordinates be transformed by $x'_h = Sx_h$ with nonsingular S . The transformed blocks satisfy $T'_{bh} = T_{bh}S^{-1}$, $T'_{hb} = ST_{hb}$, and $T'_{hh} = ST_{hh}S^{-1}$. Therefore

$$T'_{bh}(T'_{hh})^{-1}T'_{hb} = T_{bh}T_{hh}^{-1}T_{hb}. \quad (35)$$

Hence the retained effective operator (15) is invariant to arbitrary invertible coordinate changes internal to the eliminated subsystem. Hidden effective edges defined at the retained ports are therefore not artifacts of the controller-state basis.

APPENDIX B SUBGRAPH COMPOSITION VIEW

Let $T_0(s) = \text{diag}(T_1(s), \dots, T_m(s))$ contain local subgraph operators and let $C(s)$ encode port couplings. Then

$$T = T_0 - C = T_0(I - G_0C), \quad G_0 = T_0^{-1}, \quad (36)$$

so that

$$T^{-1} = (I - G_0C)^{-1}G_0. \quad (37)$$

If $\rho(G_0C) < 1$,

$$T^{-1} = G_0 + G_0CG_0 + G_0CG_0CG_0 + \dots \quad (38)$$

The terms are ordered subgraph walks. Closed products contribute to the interaction determinant $\det(I - G_0C)$, providing a direct algebraic bridge between subgraph composition and feedback cycles.

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